

Mátyás Vincze

PhD Researcher — LLM Post-Training, Reinforcement Learning, and Multi-Agent Systems
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Research & Engineering Experience

Massachusetts Institute of Technology

May 2026 – Present

Visiting PhD Researcher | Advisor: Michiel Bakker

- Developing and evaluating distillation-based post-training methods for long-horizon reasoning in language agents, combining single-agent reinforcement learning with mixtures of specialized agents on verifiable tasks.

Fondazione Bruno Kessler

Nov 2023 – Present

PhD Researcher | Advisors: Bruno Lepri and Giovanni Iacca

- Developing modular post-training methods for language models, with current work spanning self-rewarding multilingual models, multi-agent feedback, and reinforcement learning with verifiable rewards.
- Designed **SMoSE**, a sparse top-1 mixture of linear policy experts with an interpretable router. Across six MuJoCo tasks, SMoSE outperformed recent interpretable baselines on five tasks and narrowed the performance gap to black-box policies, with only 108–672 active actor parameters.
- Published the work as first author at AAAI 2025, open-sourced the reproducible PyTorch training and evaluation pipeline, and won the Continuous-Control Track of the GECCO 2024 Interpretable Control Competition.
- Supervising a bachelor's thesis extending SMoSE with critic-guided exploration annealing to improve temporal consistency in continuous-control reinforcement learning.

HK3Lab

May 2023 – Jul 2023

Research Intern | Advisor: Tommaso Furlanello

- Built multimodal lesion-classification pipelines combining whole-slide image and text embeddings using transformers and gradient boosting, achieving a 20% relative improvement in held-out accuracy over unimodal baselines in collaboration with NCTR/FDA researchers.

HSDS Lab

Mar 2022 – Jun 2022

Research Intern | Advisor: Roland Molontay

- Developed and deployed early-dropout prediction models, improving held-out accuracy by 15 percentage points; established reproducible AutoML workflows for model selection, tuning, validation, and deployment.

Vodafone

Sep 2020 – May 2021

Data Intern

- Built SQL-based KPI reports and automated Power BI dashboards, conducted ad hoc analyses, and supported the development of predictive models using large-scale telecommunications data.

Selected Publications

SMoSE – Sparse Mixture of Shallow Experts for Interpretable Reinforcement Learning in Continuous Control Tasks

Mátyás Vincze, Laura Ferrarotti, Leonardo Lucio Custode, Bruno Lepri, Giovanni Iacca

AAAI Conference on Artificial Intelligence, 2025

A Benchmark of Expert-Level Academic Questions to Assess AI Capabilities | Humanity's Last Exam

Center for AI Safety, Scale AI and HLE Contributors Consortium

Nature, 2026

Education

PhD in Computer Science

Nov 2023 – Present

University of Trento | Research conducted at Fondazione Bruno Kessler | Advisors: Bruno Lepri and Giovanni Iacca

Thesis (working title): Mixtures of Mixtures: Modular Learning for Efficient, Interpretable, and Self-Improving Agents

M.Sc. in Artificial Intelligence Systems

Sep 2021 – Oct 2023

University of Trento | Advisors: Andrea Passerini and Enver Sanginetto

Thesis: Modular Non-Autoregressive Trajectory Prediction

B.Sc. in Mathematics

Sep 2017 – Jun 2021

Budapest University of Technology and Economics | Advisor: Balázs Csanád Csáji

Thesis: Hierarchical Representation Learning Using Deep Generative Models on Controlled Complexity Data

Skills

Research Areas: LLM post-training, language agents, multi-agent systems, reinforcement learning, mixture-of-experts, interpretability, modular learning, small language models

ML Systems: PyTorch, Hugging Face Transformers/Datasets/PEFT, vLLM, Ray, veRL, FAISS, Gymnasium

Tools: Python, CUDA, NCCL, Slurm, Apptainer, Docker, Hydra, Linux, Git

Languages: Hungarian (native), English (fluent), Italian (elementary)